```
clear;clc;close all;
```

```
% 第1题
E = 105;
mu = 0.25;
t = 10;
xi = 0;
yi = 0;
xj = 10;
yj = 0;
xm = 5;
ym = 10;

ai = xj*ym - xm*yj;
aj = xm*yi - xi*ym;
am = xi*yj - xj*yi;
bi = yj - ym;
bj = ym - yi;
bm = yi - yj;
ci = - xj + xm;
cj = - xm + xi;
cm = - xi + xj;

A = (ai+aj+am) / 2;
B = 1/(2*A) * ...
    [bi,0,bj,0, bm,0 ;...
    0, ci,0, cj,0, cm;...
    ci,bi,cj,bj,cm,bm];
% 平面应力
D = E/(1-mu^2) * ...
    [1,mu,0;...
    mu,1,0;...
    0,0,(1-mu)/2];
K = B'*D*B*t*A
```

```
K = 6×6
  612.5000  175.0000 -507.5000  -35.0000 -105.00
  175.0000  350.0000   35.0000  -70.0000 -210.00
 -507.5000   35.0000  612.5000 -175.0000 -105.00
  -35.0000  -70.0000 -175.0000  350.0000  210.00
 -105.0000 -210.0000 -105.0000  210.0000  210.00
 -140.0000 -280.0000  140.0000 -280.0000
```

```
% 第2题
E = 10e6;
mu = 0.25;
t = 1;
xi = 2;
yi = 2;
xj = 4;
yj = 2;
xm = 4;
ym = 4;
```

```
ai = xj*ym - xm*yj;
aj = xm*yi - xi*ym;
am = xi*yj - xj*yi;
bi = yj - ym;
bj = ym - yi;
bm = yi - yj;
ci = - xj + xm;
cj = - xm + xi;
cm = - xi + xj;

A = (ai+aj+am) / 2;
B = 1/(2*A) * ...
    [bi,0,bj,0, bm,0 ;...
    0, ci,0, cj,0, cm;...
    ci,bi,cj,bj,cm,bm];
% 平面应变
D = E/((1+mu)*(1-2*mu)) * ...
    [1-mu,mu,0;...
    mu,1-mu,0;...
    0,0,(1-2*mu)/2];
K = B'*D*B*t*A
```

```
K = 6×6
     6000000           0    -6000000     2000000
           0     2000000     2000000    -2000000
    -6000000     2000000     8000000    -4000000
     2000000    -2000000    -4000000     8000000
           0    -2000000    -2000000     2000000
    -2000000           0     2000000    -6000000
```

```
% 第3题 a
E = 30e6;
A = 3;
L = 60;
theta = pi/4;
u1 = 0.5;
v1 = 0;
u2 = 0.25;
v2 = 0.75;

lx = cos(theta);
mx = sin(theta);
subM = [lx,mx]' * [lx,mx];
% 刚度矩阵
K = A*E/L * [subM,-subM;-subM,subM]
```

```
K = 4×4
10^5 ×
    7.5000    7.5000   -7.5000   -7.5000
    7.5000    7.5000   -7.5000   -7.5000
   -7.5000   -7.5000    7.5000    7.5000
   -7.5000   -7.5000    7.5000    7.5000
```

```
T = [lx,mx,0,0;...
    0,0,lx,mx];
% 轴向位移
d = T * [u1;v1;u2;v2]
```

```
d = 2×1
    0.3536
    0.7071
```

```
% 轴向应变
epsilon = [-1/L,1/L] * T * [u1;v1;u2;v2]
```

```
epsilon = 0.0059
```

```
% 轴向应力
sigma = E * epsilon
```

```
sigma = 1.7678e+05
```

```
% 轴向力
F = sigma * A
```

```
F = 5.3033e+05
```

```
% 第3题 b
E = 30e6;
A = 3;
u1 = 0.5;
v1 = 0;
u2 = 0.25;
v2 = 0.75;
xi = 1;
yi = 1;
xj = 4;
yj = 3;
L=sqrt((xi-xj)^2+(yi-yj)^2);
theta=atan((yj-yi)/(xj-xi));

lx = cos(theta);
mx = sin(theta);
subM = [lx,mx]' * [lx,mx];
% 刚度矩阵
K = A*E/L * [subM,-subM;-subM,subM]
```

```
K = 4×4
10^7 ×
    1.7281    1.1521   -1.7281   -1.1521
    1.1521    0.7680   -1.1521   -0.7680
   -1.7281   -1.1521    1.7281    1.1521
   -1.1521   -0.7680    1.1521    0.7680
```

```
T = [lx,mx,0,0;...
    0,0,lx,mx];
% 轴向位移
d = T * [u1;v1;u2;v2]
```

```
d = 2×1
    0.4160
    0.6240
```

```
% 轴向应变
epsilon = [-1/L,1/L] * T * [u1;v1;u2;v2]
```

```
epsilon = 0.0577
```

```
% 轴向应力
sigma = E * epsilon
```

```
sigma = 1.7308e+06
```

```
% 轴向力
F = sigma * A
```

```
F = 5.1923e+06
```